

Technical Datasheet — ArchF Drone Battery

General Information

Parameter	Value
Cell Type	Prismatic pouch cell
Model	VPF-T8R1MIMO-DSH-01
Application	Long-endurance drone
Chemistry	NMC811 / Graphite
Nominal Voltage	3.99 V (midpoint)
Nominal Capacity	5.33 Ah
Energy	19.37 Wh

Electrical Characteristics

Parameter	Min	Typical	Max	Unit
Nominal voltage	—	3.99	—	V
Charge voltage (upper cut-off)	—	4.2	—	V
Discharge voltage (lower cut-off)	—	2.5	—	V
Nominal capacity (1C)	—	5.33	—	Ah
5C discharge capacity	—	5.07	—	Ah
5C capacity retention	—	95.2	—	%
Energy density	—	466.5	—	Wh/kg
Volumetric energy density	—	857	—	Wh/L
DC resistance (1C)	—	0.115	—	mΩ
Power density	—	859	—	kW/kg

Mechanical Characteristics

Parameter	Value	Unit
Electrode height	~100	mm
Electrode width	~1027	mm
Total thickness	0.22	mm
Cell mass (formula caliber)	41.4	g
Electrolyte included	No	—

Safety Characteristics

Parameter	Value	Unit	Verdict
4C charge max temperature	361.8	K (88.6°C)	Marginal (target: ≤358K)
Lithium plating (4C charge)	No	—	PASS
Max temperature (5C discharge)	386.6	K (113.4°C)	High (drone mission uses short pulses)

Operating Conditions

Parameter	Value	Unit
Charge temperature	0 to 45	°C
Discharge temperature	-20 to 60	°C
Recommended charge rate	≤ 2C	—
Max discharge rate (continuous)	5C	—
Max discharge rate (pulse)	10C	<10s

Notes

- Mass excludes electrolyte and casing (formula-caliber mass only)
- 5C retention of 95.2% is measured relative to 1C capacity at 25°C
- Energy density computed per the contract formula: $ED = \int V \cdot I \cdot C \, dt / \Sigma(\text{layer thickness} \times (1 - \text{porosity}) \times \text{density} \times \text{area})$